

An Information-theoretic Propagation Denoising and Fusion Framework for Fake News Detection

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Abstract

Incomplete propagation data significantly hinders robust fake news detection. Recent approaches leverage large language models to simulate missing user interactions via role-playing, thereby enriching propagation with synthetic signals. However, such propagation data is intrinsically unreliable, and directly fusing it can lead to biased representations, leading to limited detection performance. In this paper, we alleviate the unreliability of synthetic propagation from the mutual information perspective and propose a novel information-theoretic propagation denoising and fusion (InfoPDF) framework to learn effective representations from both real and synthetic propagation. Specifically, we first generate attribute-specific synthetic propagation using large language models. Then we model each synthetic propagation graph as a probabilistic latent distribution to guide reliability-aware adaptive fusion with real propagation. During training, we design a mutual information-based objective to learn compressed and task-sufficient propagation representations. It jointly suppresses noisy signals across attribute-specific synthetic propagation, maintains consistency between real and synthetic propagation representations, and ensures task sufficiency for fake news detection and attribute prediction. Experiments on three real-world datasets show that InfoPDF consistently achieves superior performance across various fake news detection tasks. Further analysis demonstrates that InfoPDF can estimate attribute-level reliabilities and learn more discriminative propagation representations.

1 Introduction

The proliferation of fake news poses a significant threat to societal stability. Existing detection approaches primarily rely on content features (e.g., articles and comments) [Castillo *et al.*, 2011; Ma *et al.*, 2015; Luvembe *et al.*, 2023] or propagation patterns [Bian *et al.*, 2020; Wei *et al.*, 2021; Jiang *et al.*, 2025]. Compared to content-based methods,

propagation-based approaches generally yield superior performance by capturing rich structural features. However, real-world propagation data is often incomplete due to platform limitations or malicious deletions [Wei *et al.*, 2021; Ma *et al.*, 2022], leading to information sparsity that compromises detection accuracy. While some works attempt to enhance robustness in modeling partial propagation [Wei *et al.*, 2022; Jiang *et al.*, 2025], they remain constrained by the availability of initial real data. Recently, large language models (LLMs) have been leveraged to simulate missing interactions through attribute-mixed role-playing [Nan *et al.*, 2024; Wan *et al.*, 2024], offering a promising solution for incomplete propagation data.

However, these LLM-based simulation approaches overlook the inherent unreliability of synthetic data. Crucially, naive attribute-mixed prompting can further exacerbate this issue, i.e., LLMs may fail to fully capture the situational context of discussions, leading to generated interactions that contain hallucinations [Ji *et al.*, 2023], as well as stylistic deviations and unnatural tones [Trott *et al.*, 2023], which deviate from real-world discussions. Moreover, diverse user attributes (e.g., age, politics) impact fake news spread differently [Shu *et al.*, 2019b]. Mixing multiple attributes within a single prompt often yields inconsistent or contradictory signals. As a result, low-quality synthetic signals can contaminate informative ones, reducing the effective information available for learning and ultimately weakening detection. As shown in a preliminary analysis in Figure 1, performance varies significantly across different user attributes and simulation settings. This observation suggests that the reliability of LLM-generated synthetic propagation is attribute-dependent, motivating the explicit modeling of attribute-level reliabilities to distinguish informative signals from noise.

In this paper, we alleviate the unreliability of synthetic propagation from the mutual information perspective, and propose a novel information-theoretic propagation denoising and fusion (InfoPDF) framework to learn effective representations from both real and synthetic propagation. Specifically, InfoPDF integrates real and synthetic propagation for each attribute view to form a fused representation, which is modeled as a stochastic latent variable for uncertainty estimation. The estimated uncertainty is then converted into an attribute-level reliability score. It enables reliability-aware adaptive fusion to emphasize informative views while down-weighting

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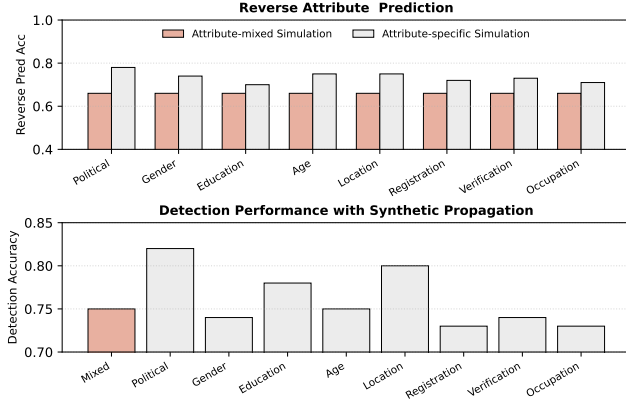


Figure 1: Preliminary analysis about different attributes and simulation settings on Twitter [Lin *et al.*, 2022]. Top: reverse attribute prediction accuracy based on the generated comments using GPT-4 Turbo. Bottom: fake news detection accuracy using synthetic propagation (GCN [Kipf and Welling, 2017] as detector).

noisy ones, thereby mitigating the negative impact of unreliable synthetic propagation on detection. During training, we design a mutual information based objective to learn compressed and task-sufficient propagation representations. It consists of three complementary objectives, i.e., minimizing the mutual information between synthetic propagation and latent representations to suppress redundant noise across different attribute views; maximizing the mutual information between real and synthetic propagation representations to reduce distributional discrepancy for propagation alignment; and maximizing the mutual information between latent representations and task labels to ensure task sufficiency for fake news detection and user attribute prediction. Through this design, InfoPDF effectively denoises and fuses multi-view synthetic propagation, improving the robustness of detection under incomplete propagation scenarios.

We conduct experiments on three public real-world fake news datasets. Experimental results demonstrate that InfoPDF consistently obtains superior performance in general detection as well as robust detection against noisy and incomplete scenarios. Further analysis reveals that our InfoPDF effectively estimates attribute-level reliability weights and extracts more discriminative propagation representations for fake news detection.

Our contributions can be summarized as follows: 1) We identify the unreliability of LLM-generated synthetic propagation for fake news detection, demonstrating that synthetic propagation can be noisy and its quality varies across attributes, which may reduce the effective information for detection. 2) We propose an LLM-generated propagation denoising and fusion framework to learn effective representations from both original and synthetic propagation. It denoise unreliable synthetic propagation via a compression-based denoising objective, and maintains the consistency between real and synthetic propagation representations as well as task sufficiency for fake news detection and user attribute prediction. 3) Experiments on three public datasets demonstrate the superiority of our proposed framework, validating its effectiveness

and robustness in handling incomplete propagation scenarios.

2 Task Formulation

Fake news detection aims to verify the authenticity of a news item, which can be formulated as a binary classification task.

Formally, each sample is represented by a graph $\mathcal{G} = (V, E)$, where $V = \{n, c_1, \dots, c_N\}$ denotes the news n and its comments, and $\mathcal{E}$ denotes explicit interactions (e.g., reply).

Due to missing interactions in practice, the observed real propagation graph is often incomplete, denoted as $G_r = (V_r, E_r)$. To enrich propagation signals, we construct K attribute-specific synthetic graphs $\{G_s^{(k)}\}_{k=1}^K$ ¹, where each $G_s^{(k)} = (V_s^{(k)}, E_s^{(k)})$ is conditioned on an attribute $a_k \in A = \{a_1, \dots, a_K\}$. The task goal can be formulated as,

$$f : (G_r, \{G_s^{(k)}\}_{k=1}^K) \longrightarrow y, \quad y \in \{0, 1\}.$$

3 Methodology

In this section, we propose a principled InfoPDF framework to learn effective representations from both real and synthetic propagation for fake news detection.

3.1 Overview of InfoPDF

The architecture of InfoPDF is shown in Figure 2.

We first leverage large language models (LLMs) to generate synthetic propagation based on each user attribute. We encode the incomplete real propagation graph and these attribute-specific synthetic propagation graphs by graph neural networks. We model the fused representation as a stochastic latent variable to estimate the reliability of synthetic propagation for each attribute, which is converted to guide the adaptive fusion where informative views are emphasized and noisy ones are down-weighted. During the training, we design a mutual-information based objective to jointly optimize the representations via minimizing the mutual information between synthetic propagation and latent representations to suppress redundant noise across different attribute views, maximizing the mutual information between real and synthetic propagation representations to reduce distributional discrepancy for propagation alignment, and maximizing the mutual information between latent representations and task labels to ensure task sufficiency for fake news detection and user attribute prediction. InfoPDF learns compressed and task-sufficient representations by exploiting both synthetic and real propagation for robust fake news detection.

3.2 Reliability-Aware Graph Fusion Network

Since the quality of synthetic propagation may vary across attributes, we model the reliability of multiple attribute-specific synthetic propagation graphs via variational estimation and use it to guide adaptive fusion.

LLM-based Synthetic Propagation Generation Given a news item, we generate an attribute-specific synthetic propagation graph $G_s^{(k)}$ for each user attribute $a_k \in A$ using a parameter-fixed LLM, following [Wan *et al.*, 2024]. Beyond

¹This allows precise assessment of each attribute’s contribution.

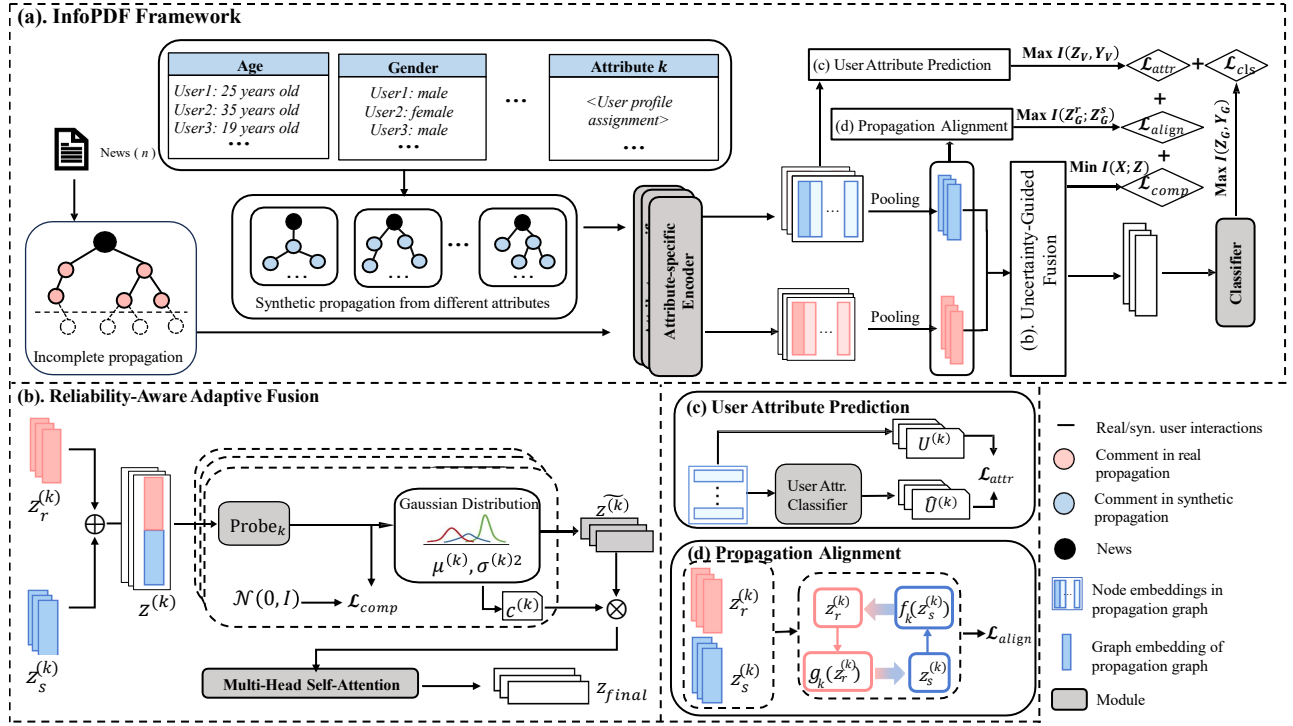


Figure 2: Overview of the proposed InfoPDF. (a) InfoPDF generates multiple attribute-specific synthetic propagation graphs and encodes them together with the incomplete real propagation. (b) InfoPDF estimates attribute-view reliability via probabilistic modeling and performs reliability-aware adaptive fusion based on the reliability-aware weighting. InfoPDF is trained with (c) user attribute prediction, (d) propagation alignment, and a Gaussian compression objective, jointly with the fake news detection loss.

them, we consider eight user attributes including age, education [Gaillard *et al.*, 2021], gender, occupation [Lim and Perrault, 2021], politics, registration time, verification status, and location [Shu *et al.*, 2019b]. The detailed generation process is provided in Appendix A.

Multi-attribute Graph-based Fusion We encode the incomplete real propagation graph G_r and each synthetic graph $G_s^{(k)}$ into latent representations. To ensure that real and synthetic views are comparable, we adopt a Siamese-style encoding strategy [Chopra *et al.*, 2005] to compute the fused representations. Specifically, for the k -th attribute, we use a graph neural network encoder denoted as $\text{Enc}_k(\cdot)$ to compute the graph-level representations of real and synthetic propagation, i.e.,

$$\mathbf{z}_r^{(k)} = \text{Pooling}(\text{Enc}_k(G_r)), \quad (1)$$

$$\mathbf{z}_s^{(k)} = \text{Pooling}(\text{Enc}_k(G_s^{(k)})). \quad (2)$$

Then we build a unified representation for each attribute view by concatenating the real and synthetic branches, i.e.,

$$\mathbf{z}^{(k)} = [\mathbf{z}_r^{(k)}; \mathbf{z}_s^{(k)}]. \quad (3)$$

To model the reliability of synthetic propagation, we apply the variational estimation to model each $\mathbf{z}^{(k)}$ as a stochastic Gaussian variable, i.e.,

$$[\boldsymbol{\mu}^{(k)}, \log \sigma^{2(k)}] = \text{Probe}_k(\mathbf{z}^{(k)}), \quad (4)$$

$$\tilde{\mathbf{z}}^{(k)} = \boldsymbol{\mu}^{(k)} + \sigma^{(k)} \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}), \quad (5)$$

where $\text{Probe}_k(\cdot)$ is a lightweight MLP that outputs the mean and log-variance of the Gaussian distribution. $\tilde{\mathbf{z}}^{(k)}$ is sampled via the reparameterization trick [Kingma and Welling, 2013]. We utilize the reliability score to compute the weight $c^{(k)}$ for fusion, i.e.,

$$c^{(k)} = \text{Softmax}\left(-\sum_{i=1}^d \log \sigma_i^{2(k)}\right), \quad (6)$$

where d is the feature dimension.

Finally, we perform reliability-aware multi-view fusion to obtain a single propagation representation for downstream fake news detection. We treat each attribute view as one token and use multi-head self-attention to capture cross-view dependencies while weighting each view by its reliability:

$$\mathbf{h}_{final} = \text{Attention}([c^{(1)}\tilde{\mathbf{z}}^{(1)}, \dots, c^{(K)}\tilde{\mathbf{z}}^{(K)}]). \quad (7)$$

Fake News Detection The final prediction of fake news is computed by:

$$\hat{y} = \text{Softmax}(\mathbf{W}_{cls}\mathbf{h}_{final} + \mathbf{b}_{cls}), \quad (8)$$

where $\mathbf{W}_{cls}$ and $\mathbf{b}_{cls}$ are learnable parameters.

3.3 Training Objectives based on Mutual Information

We design a mutual information based objective to learn compressed and task-sufficient propagation representations.

Our goal is to (i) minimize the mutual information between the input of synthetic propagation and latent representations to suppress redundant noise across different attribute views for redundancy compression, (ii) maximize the mutual information between real and synthetic propagation representations to reduce distributional discrepancy for propagation alignment, and (iii) maximize the mutual information between latent representations and task labels to ensure task sufficiency for both fake news detection and user attribute prediction.

Formally, let X denote the input graph (i.e., the unified view of real and synthetic propagation), Y is the ground-truth task label (i.e., graph-level label Y_G for fake news detection or node-level label Y_V for user attribution label), node-level and graph-level representations are Z_V and Z_G . Given the Markov chain constraint $(Y_G, Y_V) \rightarrow X \rightarrow Z_V \rightarrow Z_G$, the overall objective of InfoPDF is formulated as:

$$\max \underbrace{I(Z_G; Y_G) + \alpha I(Z_V; Y_V)}_{\text{Task Sufficiency}} - \underbrace{\gamma I(X; Z_G)}_{\text{Compression}} + \underbrace{\lambda I(Z_G^r; Z_G^s)}_{\text{Alignment}}, \quad (9)$$

where Z_G^r and Z_G^s are graph-level representations learned from real and synthetic propagation. α , λ and γ are trade-off hyperparameters. controlling the weights of task supervision, propagation alignment, and information compression.

Maximizing $I(Z_G; Y_G)$ and $I(Z_V; Y_V)$ We jointly optimize fake news detection and user attribute prediction via supervised learning objectives, which serve as variational lower bounds of the corresponding mutual information terms $I(Z_G; Y_G)$ and $I(Z_V; Y_V)$. First, maximizing $I(Z_G; Y_G)$ encourages the learned graph-level representation Z_G to be maximally informative for fake news classification task. In practice, this objective is implemented by minimizing the binary cross-entropy loss,

$$\mathcal{L}_{cls} = -\log p_{\theta_{cls}}(Y_G | Z_G), \quad (10)$$

where θ_{cls} refers to trainable parameters for fake news classifier. Minimizing $\mathcal{L}_{cls}$ corresponds to maximizing the conditional log-likelihood $\log p(Y_G | Z_G)$ and thus a lower bound of $I(Z_G; Y_G)$.

Additionally, to encourage node-level representations to retain attribute-discriminative information, we introduce an auxiliary node-level attribute prediction objective to maximize $I(Z_V; Y_V)$. We minimize the following negative log-likelihood loss, i.e.,

$$\mathcal{L}_{attr} = \frac{1}{K} \sum_{k=1}^K \frac{1}{\log |\mathcal{C}_k|} \left(-\log p_{\theta_{attr}^{(k)}}(Y_V^{(k)} | Z_V^{(k)}) \right) \quad (11)$$

where $|\mathcal{C}_k|$ is the number of classes for attribute a_k . $\theta_{attr}^{(k)}$ refers to trainable parameters for the k -th attribution classifier. We use the log-scale factor to balance attribute spaces with different labels. Minimizing $\mathcal{L}_{attr}$ maximizes the conditional likelihood of attribute labels given node representations, serving as a variational lower bound for $I(Z_V; Y_V)$.

Minimizing $I(X; Z)$ To suppress redundant information introduced by LLM-generated synthetic propagation, we adopt the variational information bottleneck [Alemi *et al.*,

2017] to minimize the mutual information between the latent representation and the input view. Specifically, we approximate the graph-level distribution with a variational Gaussian posterior $q(\tilde{\mathbf{z}}^{(k)} | \mathbf{z}^{(k)}) = \mathcal{N}(\boldsymbol{\mu}^{(k)}, \boldsymbol{\sigma}^{2(k)})$, and impose a standard normal prior $p(\tilde{\mathbf{z}}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$. The mutual information $I(X; Z)$ can be upper bounded by the Kullback–Leibler (KL) divergence between the posterior and the prior. That is, we minimize the following compression loss,

$$\mathcal{L}_{comp} = \sum_{k=1}^K D_{KL}(q(\tilde{\mathbf{z}}^{(k)} | \mathbf{z}^{(k)}) \| p(\tilde{\mathbf{z}})). \quad (12)$$

Minimizing $\mathcal{L}_{comp}$ reduces the upper bound of $I(X; Z)$, encouraging the representations to suppress task-irrelevant noise of LLM-generated synthetic propagation.

Maximizing $I(Z_G^r; Z_G^s)$ Since directly estimating mutual information is intractable, we adopt a deterministic alignment objective as a surrogate. Specifically, we introduce a bi-directional alignment loss function to reduce the gap between real and synthetic propagation representations. The intuition is that when two views share high mutual information, one view should be predictive of the other [Yu *et al.*, 2025]. For each attribute view k , we employ two lightweight predictors $f_k(\cdot)$ and $g_k(\cdot)$ (implemented as MLPs) to reconstruct one representation from the other. The loss function is defined as,

$$\mathcal{L}_{align} = \sum_{k=1}^K \left(\|f_k(\mathbf{z}_s^{(k)}) - \mathbf{z}_r^{(k)}\|_2^2 + \|g_k(\mathbf{z}_r^{(k)}) - \mathbf{z}_s^{(k)}\|_2^2 \right). \quad (13)$$

Totally, the overall training loss of InfoPDF is defined as,

$$\mathcal{L} = \mathcal{L}_{cls} + \alpha \mathcal{L}_{attr} + \lambda \mathcal{L}_{align} + \gamma \mathcal{L}_{comp}. \quad (14)$$

where α , λ , and γ are trade-off hyperparameters.

3.4 LLM-free Inference

To enable efficient deployment, we also provide an LLM-free mode of InfoPDF without online simulation. We reconstruct the synthetic attribute representation from the real representation using the trained predictor g_k :

$$\hat{\mathbf{z}}_s^{(k)} = g_k(\mathbf{z}_r^{(k)}). \quad (15)$$

We then form $\hat{\mathbf{z}}^{(k)} = [\mathbf{z}_r^{(k)}; \hat{\mathbf{z}}_s^{(k)}]$ and feed it into the reliability-aware fusion to obtain h_{final} for fake news detection. This mode preserves the benefits of multi-view fusion while avoiding the computational overhead of online LLM-based propagation simulation.

4 Experiments

4.1 Experimental Setups

Datasets We conduct experiments on three public datasets, i.e., Twitter [Lin *et al.*, 2022], CED [Song *et al.*, 2019] and PHEME [Zubiaga *et al.*, 2016]. Following previous works [Chen *et al.*, 2025], we divide the datasets into training, validation, and testing sets in a ratio of 7:1:2. The details of the datasets are provided in Appendix B.

	Method	Twitter			CED			PHEME		
		Acc	Macro-F1	AUC	Acc	Macro-F1	AUC	Acc	Macro-F1	AUC
Content-based	BERT	71.12±2.43	70.86±2.71	71.13±2.38	88.44±0.96	88.43±0.89	88.48±0.93	81.83±1.27	79.22±1.64	78.88±1.51
	DeBerta	75.43±1.92	74.47±1.87	73.10±2.14	89.97±0.73	89.32±0.79	93.30±0.65	82.54±1.09	80.24±1.21	80.65±1.08
	EANN	72.84±1.68	72.41±1.79	72.74±1.83	89.73±0.76	89.69±0.72	89.63±0.74	82.76±0.89	80.79±1.02	81.20±0.94
	dEFEND	75.12±1.41	73.56±1.47	75.13±1.39	90.12±0.63	89.98±0.71	96.57±0.49	83.24±0.84	82.45±0.91	89.42±0.72
	DualEmo	73.27±1.57	73.01±1.66	73.28±1.61	90.16±0.69	90.29±0.73	96.17±0.56	83.60±0.88	83.34±0.86	91.28±0.74
	LLM _{text}	56.00±4.23	54.60±4.41	57.24±3.96	48.08±2.53	37.22±2.74	45.47±2.48	33.74±3.35	28.64±3.52	31.78±3.41
	ARG	77.91±0.98	68.50±1.31	68.32±1.23	92.92±0.31	92.88±0.35	97.59±0.26	85.62±0.49	84.10±0.54	91.89±0.42
	LLM _{comments}	60.75±3.64	60.47±3.78	61.27±3.52	80.10±1.41	80.05±1.38	83.63±1.24	53.51±2.41	52.77±2.58	54.18±2.33
	TELLER	79.35±0.86	77.82±0.91	82.46±0.75	92.48±0.26	92.31±0.29	97.26±0.23	84.78±0.47	83.95±0.51	90.52±0.44
	GenFEND									
	w/ BERT	78.26±1.38	75.64±1.52	74.78±1.61	90.23±0.59	90.04±0.63	93.84±0.52	83.84±0.81	81.23±0.87	87.89±0.73
	w/ dEFEND	82.25±0.74	82.04±0.81	90.70±0.68	93.23±0.24	92.96±0.27	98.14±0.16	85.07±0.39	84.66±0.45	91.63±0.36
	CAMERED	82.85±0.65	82.52±0.71	91.15±0.58	93.58±0.22	93.25±0.25	98.26±0.15	85.44±0.36	85.02±0.42	91.95±0.33
Propagation-based	GCN	78.02±2.91	77.67±3.08	86.61±2.62	91.59±1.12	91.17±1.08	97.62±0.79	80.59±0.30	75.55±1.05	86.96±1.52
	GAT	79.31±2.73	78.86±2.87	87.09±2.46	88.23±1.29	87.68±1.31	97.46±0.82	80.41±0.46	77.13±1.25	86.69±0.35
	GraphSAGE	81.03±2.58	79.75±2.69	88.39±2.34	89.26±1.19	88.92±1.26	97.75±0.75	79.03±1.29	75.70±0.52	86.57±0.24
	BiGCN	82.76±1.56	82.71±1.49	90.67±1.34	91.74±0.69	91.63±0.74	96.64±0.59	82.70±0.35	80.28±0.88	88.44±0.42
	UPFD	79.74±1.89	79.02±2.01	89.48±1.71	88.79±1.02	88.38±0.98	96.66±0.71	81.05±0.41	77.53±1.12	87.34±0.65
	EBGCN	83.19±0.91	82.60±0.98	91.32±0.83	93.96±0.36	93.76±0.39	98.57±0.23	83.89±0.28	80.72±0.52	89.14±0.31
	UPSR	83.62±0.87	83.22±0.94	90.88±0.81	93.36±0.33	92.96±0.36	97.97±0.26	82.95±0.31	82.95±0.37	89.70±0.28
	RAGCL	84.42±0.72	84.15±0.78	91.76±0.64	94.10±0.29	93.95±0.32	98.42±0.19	84.24±0.25	83.58±0.41	90.35±0.22
	LLM _{propagation}	48.84±4.42	46.76±4.59	47.72±4.31	71.15±1.91	67.32±2.04	73.78±1.78	52.14±2.64	52.04±2.78	55.62±2.52
	DELL									
	- single	80.17±1.62	79.75±1.54	88.39±1.38	92.12±0.71	91.69±0.68	96.66±0.54	82.60±0.38	80.75±0.73	88.89±0.46
	- vanilla	80.17±1.47	82.55±1.34	90.75±1.21	92.23±0.62	91.75±0.66	98.07±0.45	82.52±0.42	80.60±0.68	89.20±0.39
	- confidence	78.97±1.69	78.10±1.73	90.21±1.44	91.03±0.68	90.10±0.72	97.79±0.49	83.03±0.45	81.08±0.76	89.07±0.41
	- selective	81.22±1.43	81.02±1.51	86.79±1.59	92.28±0.59	91.32±0.65	97.37±0.48	83.29±0.39	82.75±0.62	89.02±0.37
	EIN	84.33±0.64	83.97±0.71	91.25±0.58	94.25±0.26	94.08±0.29	98.35±0.17	85.27±0.23	84.46±0.38	90.88±0.21
	InfoPDF (Ours)									
	- GCN	85.63*±0.39	85.32*±0.40	91.45±0.32	96.36*±0.07	96.24*±0.06	98.98±0.05	87.65*±0.24	85.47±0.26	91.96±0.85
	- GAT	85.43*±0.32	85.15*±0.34	91.42±0.22	97.40* ±0.18	97.28* ±0.18	99.31*±0.04	88.17* ±0.21	85.96*±0.28	92.88*±0.19
	- GraphSAGE	86.90* ±0.34	86.51* ±0.35	93.25* ±0.27	97.25*±0.18	97.13*±0.19	99.33* ±0.04	88.11*±0.22	86.11* ±0.25	93.12* ±0.42
	InfoPDF-Inference									
	- GCN	84.78±0.43	84.45±0.46	91.85±0.38	94.48±0.12	94.31±0.13	98.52±0.08	85.42±0.28	84.68±0.31	91.12±0.52
	- GAT	84.62±0.36	84.28±0.39	91.68±0.31	94.55±0.21	94.38±0.22	98.48±0.06	85.58±0.24	84.85±0.27	91.28±0.21
	- GraphSAGE	84.95±0.38	84.72±0.41	92.18±0.33	94.71±0.20	94.56±0.21	98.61±0.06	85.71±0.25	84.92±0.28	91.45±0.38

Table 1: Results (%) for fake news detection on three datasets. The best results are in **boldface**. For each method, we run it five times and report the average results with standard deviations. InfoPDF-Inference represents the LLM-free inference mode of InfoPDF. * represents statistical significance over state-of-the-art baselines under the t -test ($p < 0.05$).

Comparison Methods We compare InfoPDF with 11 content-based methods including BERT [Devlin *et al.*, 2018] and DeBERTa [He *et al.*, 2020], EANN [Wang *et al.*, 2018], dEFEND [Shu *et al.*, 2019a], DualEmo [Zhang *et al.*, 2021], ARG [Hu *et al.*, 2024], GeFEND [Nan *et al.*, 2024], TELLER [Liu *et al.*, 2024a], CAMERED [Wang *et al.*, 2025], LLM_{text}, and LLM_{comment} [Chen *et al.*, 2025], and 11 propagation-based methods including GCN [Kipf and Welling, 2017], GAT [Veličković *et al.*, 2018], GraphSAGE [Hamilton *et al.*, 2017], Bi-GCN [Bian *et al.*, 2020], EBGCN [Wei *et al.*, 2021], UPSR [Wei *et al.*, 2022], RAGCL [Cui and Jia, 2024], UPFD [Dou *et al.*, 2021], DELL [Wan *et al.*, 2024], EIN [Jiang *et al.*, 2025], and LLM_{propagation} [Chen *et al.*, 2025]. Details of baselines are listed in Appendix B.

Evaluation Metrics We evaluate with three classification metrics, including the accuracy (Acc), macro-averaged F1 score (Macro-F1), and area under the ROC curve (AUC).

Implementation Details All experiments are implemented on a single NVIDIA Tesla V100 GPU. We utilize the Adam optimizer with a learning rate of $2e-4$, a batch size of 32, and

a hidden dimension of 128. The size of synthetic propagation graphs is fixed to 30 nodes (1 root news node and 29 comment nodes). For LLM-based backbones, we employ *Mistral-7B-Instruct-v0.2* for Twitter, PHEME, and *Qwen1.5-7B-Chat* for CED. More implementation details are listed in Appendix B.

4.2 Main Results

We evaluate the effectiveness of InfoPDF across different detection scenarios.

General Detection Table 1 reports the performance comparison on three public datasets. Overall, InfoPDF consistently achieves the best results across different GNN backbones and evaluation metrics, demonstrating its effectiveness and strong generalization under incomplete propagation. For instance, InfoPDF improves the GraphSAGE backbone by +5.87% accuracy scores on Twitter. Moreover, LLM-free variant InfoPDF-Inference consistently outperforms comparison methods while remaining competitive with the full InfoPDF model, enabling efficient LLM-free deployment. This validates that InfoPDF can distill the benefit of LLM-generated synthetic propagation into an efficient LLM-free

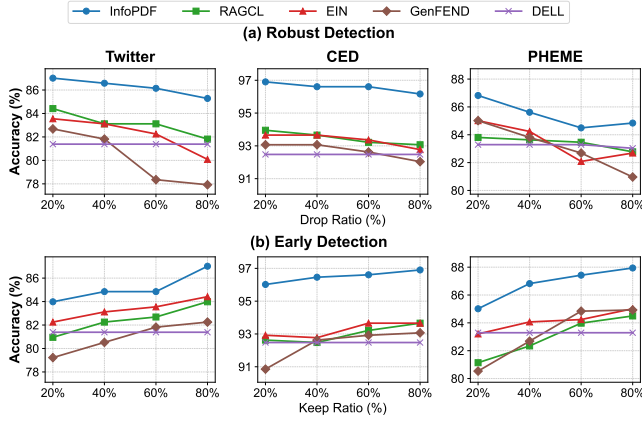


Figure 3: Results (%) of robust and early fake news detection on two incomplete propagation scenarios.

Method	Twitter		CED		PHEME	
	Acc	Macro-F1	Acc	Macro-F1	Acc	Macro-F1
InfoPDF	86.90	86.51	97.25	97.13	88.11	86.11
w/o RP	84.34	84.10	94.05	93.89	86.21	85.97
w/o Attr	84.70	84.23	94.79	94.54	86.04	85.64
w/o RW	83.84	83.33	94.39	94.25	85.78	85.14
w/o Attention	84.48	84.14	94.54	94.36	85.86	85.24
w/o $\mathcal{L}_{align}$	83.20	82.93	93.95	93.85	85.34	84.91
w/o $\mathcal{L}_{comp}$	84.42	84.17	94.40	94.16	85.87	85.48
w/o $\mathcal{L}_{attr}$	84.99	84.83	94.69	94.32	86.12	85.78

Table 2: Ablation results on three datasets. The best performance is highlighted in bold. We apply GraphSAGE as the GNN backbone.

detector, making it more practical for real-world deployment.

Robust Detection Figure 3(a) shows the results of InfoPDF and baselines for robust detection. Specifically, we simulate noisy propagation settings by randomly removing edges and masking node features with ratios of 20%, 40%, 60%, and 80%. From results, InfoPDF shows smaller performance degradation, exhibiting stronger robustness towards different incomplete propagation settings.

Early Detection Figure 3(b) shows the results for early detection on three datasets. We preserve the temporal order of propagation and only using partial propagation observed at early time steps for prediction. Our InfoPDF achieves competitive performance even when only limited early propagation is available, showing its effectiveness on early detection.

4.3 Ablation Study

We conduct ablation studies to examine the contribution of each component in InfoPDF. The ablation results are shown in Table 2.

For propagation modeling for fake news detection, removing the incomplete real propagation (w/o RP) consistently degrades performance, showing the necessary of real propagation data. Replacing attribute-specific simulation with attribute-mixed generation (w/o Attr) leads to clear drops, indicating that modeling attributes separately yields more reliable synthetic signals. Moreover, removing either reliability-aware weighting (w/o RW) or multi-head self-attention fusion

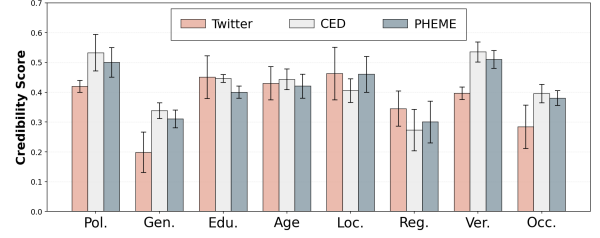


Figure 4: Attribute-level credibility distributions with user variability across three datasets.

Dataset	Method	Supervised Metrics			Unsupervised Metrics		
		ARI $\uparrow$	NMI $\uparrow$	FMI $\uparrow$	SC $\uparrow$	CHI $\uparrow$	DBI $\downarrow$
Twitter	RP	30.14	23.51	65.08	0.24	85.30	1.52
	Direct Fusion	39.34	30.87	69.56	0.23	77.60	1.59
	InfoPDF	43.82	34.67	71.79	0.56	443.54	0.59
CED	RP	65.28	57.60	82.70	0.56	1887.89	0.50
	Direct Fusion	60.13	50.60	80.10	0.38	543.31	1.06
	InfoPDF	89.09	81.56	94.54	0.67	2564.49	0.43
PHEME	RP	41.47	28.77	74.58	0.11	139.66	2.53
	Direct Fusion	40.36	28.71	73.07	0.10	133.82	2.77
	InfoPDF	51.12	39.52	77.64	0.49	1699.84	0.74

Table 3: Representation evaluation results (%) against the propagation data from different strategies. Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), and Fowlkes-Mallows Index (FMI) evaluate the accuracy of clustering. Silhouette Coefficient (SC), Calinski-Harabasz Index (CHI), and Davies-Bouldin Index (DBI) evaluate the separation and compactness of clustering. SC, CHI, and DBI are evaluated based on K-Means, and we define the number of clusters K as the true number of categories, i.e., K = 2 for Twitter, PHEME and CED.

causes performance degradation, verifying their effectiveness for detection.

For ablation variants of training objectives, removing the propagation alignment objective (w/o $\mathcal{L}_{align}$) results in the largest performance drop, highlighting the importance of reducing the representation gap between real and synthetic propagation. Removing the compression regularization (w/o $\mathcal{L}_{comp}$) also consistently harms performance, showing that the compression helps filter redundant features in representations. Finally, removing attribute supervision (w/o $\mathcal{L}_{attr}$) leads to further degradation, demonstrating that preserving the sufficiency of representations for user attributes is beneficial for detection.

4.4 Further Analysis

We provide more analysis on (i) reliability of synthetic propagation, (ii) representation quality under different fusion strategies, and (iii) hyperparameter sensitivity. Additionally, we analyze the effect of different synthetic propagation size, semantics of synthetic propagation, representation visualization, and case study in Appendix C.

Reliability Analysis of Synthetic Propagation Figure 4 visualizes the attribute-view reliability scores estimated by InfoPDF across three datasets. Overall, we observe clear differences in the estimated reliability across attribute views, indicating that LLM-generated propagation signals exhibit varying levels of consistency and usefulness. Moreover, the

Method	Twitter		CED		PHEME	
	Acc	Macro-F1	Acc	Macro-F1	Acc	Macro-F1
InfoPDF (Gaussian)						
w/ GCN	85.63	85.32	96.36	96.24	87.65	85.74
w/ GAT	85.43	85.15	97.40	97.28	88.17	85.96
w/ SAGE	86.90	86.51	97.25	97.13	88.11	86.11
InfoPDF (Beta)						
w/ GCN	85.34	85.01	96.08	95.94	87.68	85.36
w/ GAT	85.35	85.04	95.89	95.74	87.93	85.61
w/ SAGE	85.34	85.03	96.17	96.01	87.41	85.15

Table 4: Comparison results (%) of InfoPDF using different distributions for reliability estimation. Best results are in **bold**.

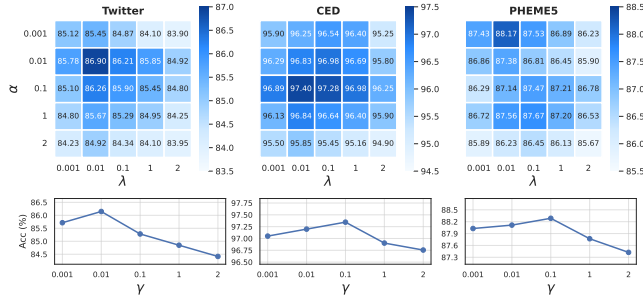


Figure 5: Performance analysis with varying α , λ and γ parameters. We use accuracy score (%) for evaluation.

reliability scores also vary across instances within the same attribute view, suggesting that InfoPDF performs instance-wise reliability estimation rather than relying on static attribute-level priors. These results support the effectiveness of reliability-aware fusion in adaptively leveraging informative synthetic views.

Representation Quality Analysis We analyze the quality of representations learned by InfoPDF and several variants including: 1) RP, which encodes only the incomplete real propagation; and 2) Direct Fusion, which fuses the real propagation representation with multiple attribute-specific synthetic representations directly. Following existing studies [Hu *et al.*, 2023], we compute supervised clustering metrics (ARI, NMI, FMI) and unsupervised clustering metrics (SC, CHI, DBI) to evaluate the representation quality. Table 3 reports results of our method and comparison variants. The more accurate supervised clustering results demonstrate the class discrimination capability of InfoPDF. Better results in terms of unsupervised metrics indicate the high structural quality of representations learned by our method, reflected in compact and well-separated clustering patterns.

Comparison with Different Reliability Estimation We study the impact of probabilistic modeling choices for reliability estimation by comparing Gaussian and Beta distributions [Ma *et al.*, 2024]. As shown in Table 4, Gaussian-based InfoPDF consistently achieves better performance across datasets. This suggests that Gaussian distribution modeling provides a more suitable way for reliability estimation as they better align with the continuous and unbounded nature of the propagation representation space.

Hyperparameter Analysis We analyze the sensitivity of InfoPDF to key trade-off hyperparameters α , λ , and γ during training, as shown in Figure 5. A small λ (e.g., 0.01) con-

sistently yields strong performance, indicating that mild alignment suffices to bridge the real-synthetic gap. The optimal α varies across datasets, reflecting distinct underlying attribute distributions and propagation patterns. For γ , moderate values achieve the best balance between denoising and information preservation, whereas excessive compression risks suppressing task-relevant signals. Overall, these results demonstrate that InfoPDF is robust to hyperparameter variations and can be reliably tuned in practice.

5 Related Work

Fake News Detection Existing methods primarily focus on content and propagation. Content-based methods extract semantic patterns via feature engineering [Castillo *et al.*, 2011; Ma *et al.*, 2015], deep neural networks [Ruchansky *et al.*, 2017; Yu *et al.*, 2017], and PLMs [Kaliyar *et al.*, 2021]. Auxiliary tasks like sentiment analysis [Luvembe *et al.*, 2023] and comment utilization [Shu *et al.*, 2019a] are also employed to complement textual features. Propagation-based methods model dissemination structures using time series [Liu and fang Brook Wu, 2018] or graphs [Bian *et al.*, 2020; Wei *et al.*, 2021; Wei *et al.*, 2022], sometimes incorporating multi-relational interactions [Dou *et al.*, 2021]. However, incomplete propagation remains a challenge [Wei *et al.*, 2021; Ma *et al.*, 2022]. While contrastive learning [Ma *et al.*, 2022; Cui and Jia, 2024] and uncertainty modeling [Wei *et al.*, 2021; Wei *et al.*, 2022] improve robustness, they fundamentally rely on the availability of initial real propagation, which is often inaccessible in early detection scenarios.

LLM-based Propagation Generation LLMs demonstrate significant potential in simulating human behavior and social knowledge [Liu *et al.*, 2024b; Wang *et al.*, 2025], having recently been established as a verifiable and scalable research method for capturing complex social dynamics [Anthis *et al.*, 2025]. Studies on fake news detection have moved beyond simple discussion simulation [Nan *et al.*, 2024; Wan *et al.*, 2024] to modeling intricate dissemination behaviors such as liking, sharing, and content regulation within multi-agent social environments [Liu *et al.*, 2025]. However, these methods often overlook the inherent unreliability risks in synthetic propagation. This paper mitigates the unreliability in LLM-generated propagation from the mutual information perspective.

6 Conclusion

This paper alleviates the unreliability in LLM-generated propagation from the mutual information perspective. We propose a novel InfoPDF to learn effective representations from both real and multiple LLM-generated synthetic propagation for fake news detection. InfoPDF denoises unreliable synthetic propagation via variational information bottleneck, and maintains the consistency between real and synthetic propagation representation as well as task sufficiency to fake news detection and user attribute prediction. Experiments on three datasets demonstrate that InfoPDF achieves superior performance and robustness in data-sparse scenarios, while its LLM-free inference mode offers a practical solution for efficient deployment.

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